# DAX & Data Visualization

## DAX Calculations:
## Calculated Columns

● **Create a Calculated Column for 'Category Type':** Add a calculated column in the Order Details table that combines the 'Category' and 'Sub-Category' columns into a single 'Category Type' column.

```
1 Category Type = CONCATENATE('Order Details'[Category],"-"&'Order Details'[Sub-Category])
```

| Order ID | Amount | Profit | Quantity | Category | Sub-Category | Profit Margin | Profit Status | Category Type |
|---|---|---|---|---|---|---|---|---|
| B-25604 | 65 | 17 | 2 | Clothing | T-shirt | 26.15% | Profit | Clothing-T-shirt |
| B-25606 | 87 | 4 | 2 | Clothing | Shirt | 4.60% | Profit | Clothing-Shirt |
| B-25607 | 50 | 15 | 4 | Clothing | Leggings | 30.00% | Profit | Clothing-Leggings |
| B-25615 | 68 | 20 | 5 | Clothing | Hankerchief | 29.41% | Profit | Clothing-Hankerchief |
| B-25616 | 42 | 12 | 5 | Clothing | Hankerchief | 28.57% | Profit | Clothing-Hankerchief |
| B-25619 | 353 | 90 | 8 | Clothing | Saree | 25.50% | Profit | Clothing-Saree |
| B-25623 | 53 | 1 | 4 | Clothing | Stole | 1.89% | Profit | Clothing-Stole |

● **Calculate Revenue per Order in Order Details Table:** Create a calculated column in the Order Details table to compute the revenue (Amount * Quantity) per order.

```
1 Revenue = ('Order Details'[Amount]*'Order Details'[Quantity])
```

| Order ID | Amount | Profit | Quantity | Category | Sub-Category | Profit Margin | Profit Status | Category Type | Revenue |
|---|---|---|---|---|---|---|---|---|---|
| B-25604 | 65 | 17 | 2 | Clothing | T-shirt | 26.15% | Profit | Clothing-T-shirt | 130 |
| B-25606 | 87 | 4 | 2 | Clothing | Shirt | 4.60% | Profit | Clothing-Shirt | 174 |
| B-25607 | 50 | 15 | 4 | Clothing | Leggings | 30.00% | Profit | Clothing-Leggings | 200 |
| B-25615 | 68 | 20 | 5 | Clothing | Hankerchief | 29.41% | Profit | Clothing-Hankerchief | 340 |
| B-25616 | 42 | 12 | 5 | Clothing | Hankerchief | 28.57% | Profit | Clothing-Hankerchief | 210 |
| B-25619 | 353 | 90 | 8 | Clothing | Saree | 25.50% | Profit | Clothing-Saree | 2824 |
| B-25623 | 53 | 1 | 4 | Clothing | Stole | 1.89% | Profit | Clothing-Stole | 212 |

● **Create a Calculated Column to Categorize Sales:** Add a calculated column named ‘Sales Category’ in the Order Details table that categorizes each order as 'Above Average' or 'Below Average' based on the Amount value.

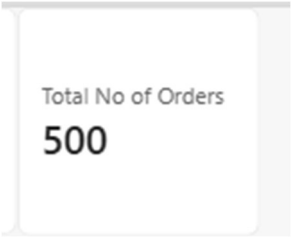
```
1 Categorize Sales = IF('Order Details'[Amount]<=270,"Below Average","Above Average")
```

| Order ID | Amount | Profit | Quantity | Category | Sub-Category | Profit Margin | Profit Status | Category Type | Revenue | Categorize Sales |
|---|---|---|---|---|---|---|---|---|---|---|
| B-25604 | 65 | 17 | 2 | Clothing | T-shirt | 26.15% | Profit | Clothing-T-shirt | 130 | Below Average |
| B-25606 | 87 | 4 | 2 | Clothing | Shirt | 4.60% | Profit | Clothing-Shirt | 174 | Below Average |
| B-25607 | 50 | 15 | 4 | Clothing | Leggings | 30.00% | Profit | Clothing-Leggings | 200 | Below Average |
| B-25615 | 68 | 20 | 5 | Clothing | Hankerchief | 29.41% | Profit | Clothing-Hankerchief | 340 | Below Average |
| B-25616 | 42 | 12 | 5 | Clothing | Hankerchief | 28.57% | Profit | Clothing-Hankerchief | 210 | Below Average |
| B-25619 | 353 | 90 | 8 | Clothing | Saree | 25.50% | Profit | Clothing-Saree | 2824 | Above Average |
| B-25623 | 53 | 1 | 4 | Clothing | Stole | 1.89% | Profit | Clothing-Stole | 212 | Below Average |
| B-25624 | 26 | 12 | 3 | Clothing | Hankerchief | 46.15% | Profit | Clothing-Hankerchief | 78 | Below Average |

## Calculated Measures

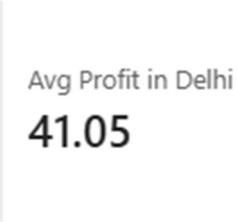
1. **Order Count-** Create a measure that calculates the **total number of orders** in the **Order Details** table.

```
Total No of Orders = COUNTA('Order Details'[Order ID])
```



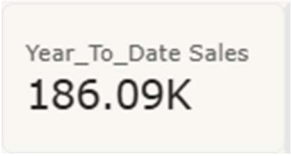

2. **Average Profit in Delhi-** Create a measure that calculates the **average profit for orders placed in Delhi**.

```
 Avg Profit in Delhi = CALCULATE(AVERAGE('Order
Details'[Profit]),FILTER('List of Orders','List of
Orders'[State]="Delhi"))
```



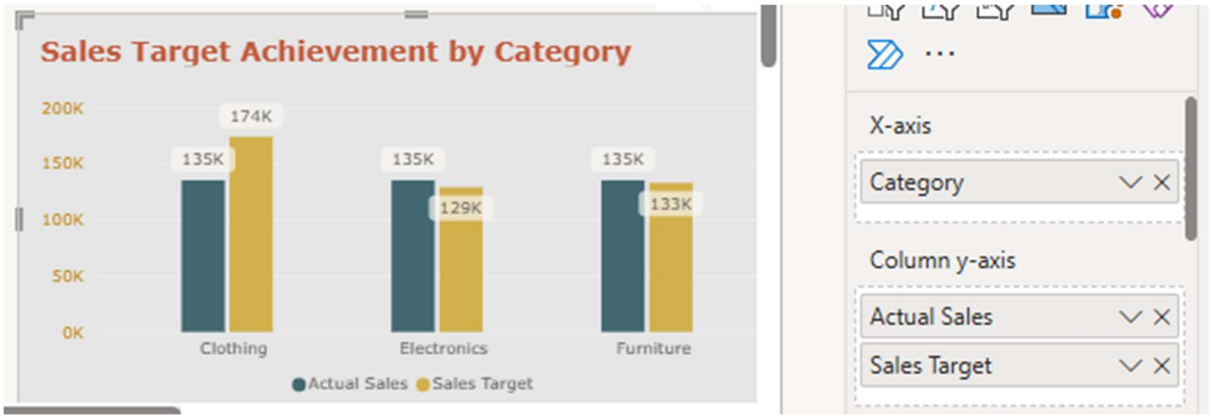

3. **Year-to-Date (YTD) Sales-** Create a measure that calculates **total sales accumulated from the beginning of the year up to the current order date**.

```
Year_To_Date Sales = TOTALYTD(sum('Order Details'[Revenue]),'List of
Orders'[Order Date].[Date])
```

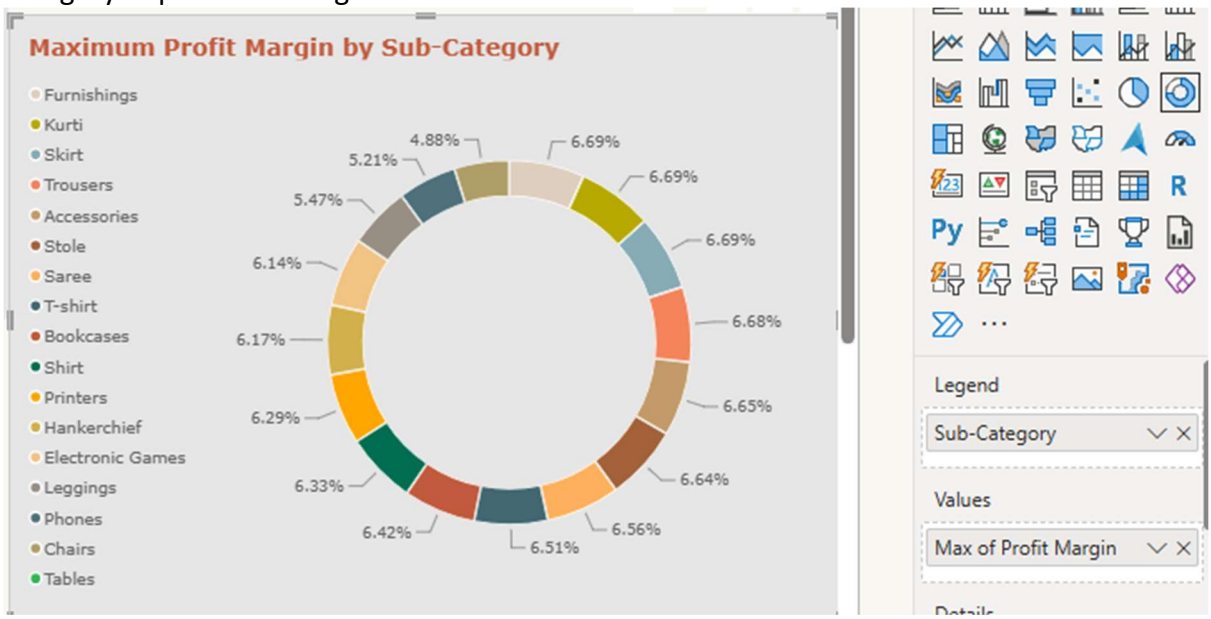


## Data Visualization

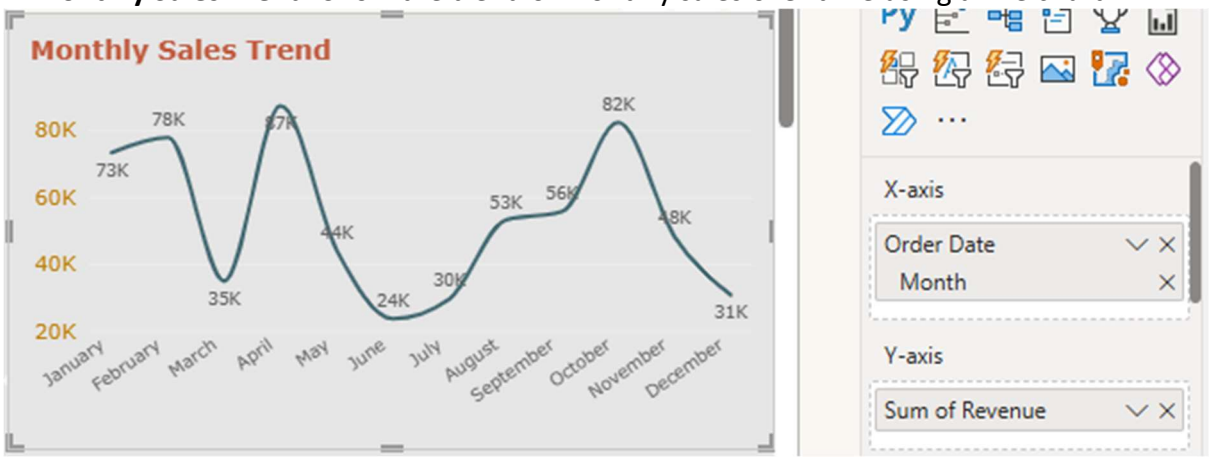
● **Sales Target Achievement by Category:** Compare actual sales with sales targets by category using a clustered column chart.



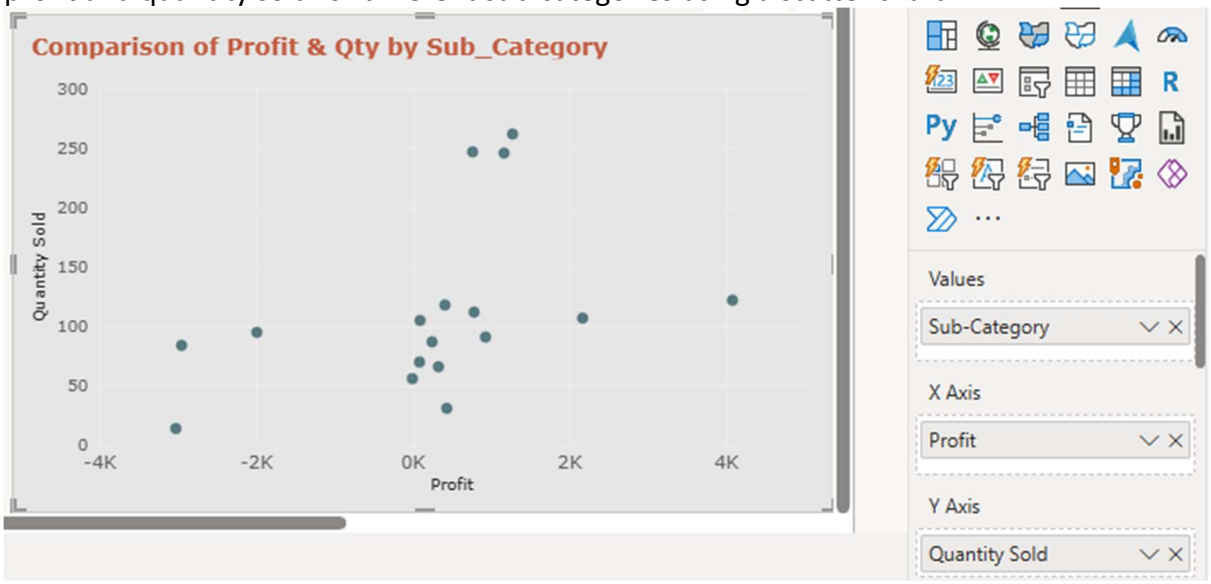

● **Max Profit Margin by Sub-Category:** Analyze the maximum profit margin for each sub-category of products using a donut chart.



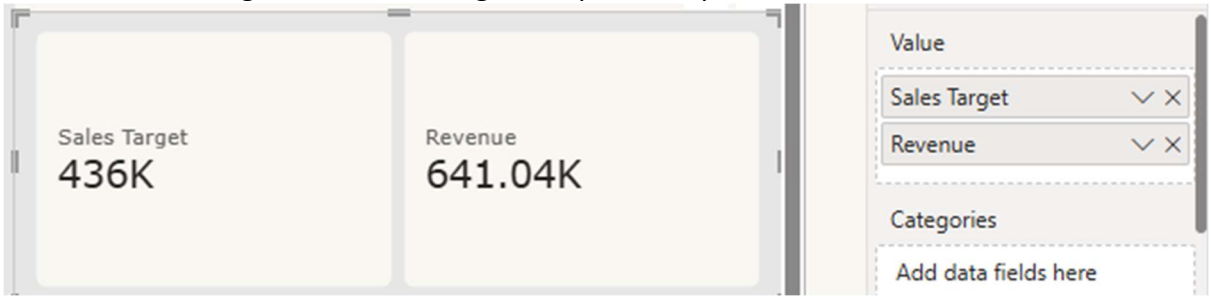
● **Monthly Sales Trend:** Show the trend of monthly sales over time using a line chart.



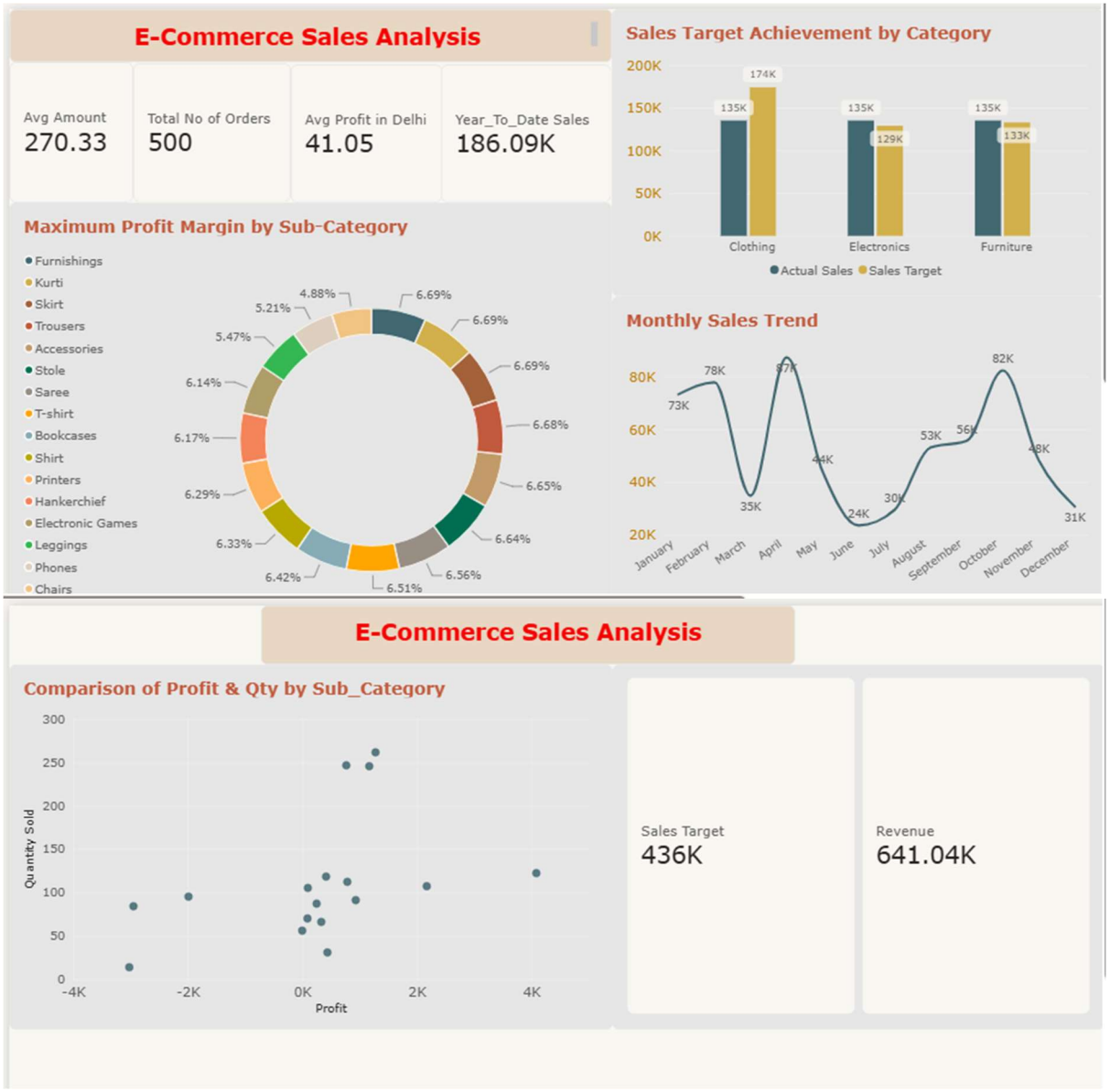

● **Comparison of Profit and Quantity by Sub-Category:** Compare the relationship between profit and quantity sold for different sub-categories using a scatter chart.




● **Comparison of Total Sales Amount and Target:** Create cards to succinctly display the total sales amount alongside the sales target for quick comparison



# Dashboard


E-Commerce Sales Analysis
Avg Amount
270.33
Total No of Orders
500
Avg Profit in Delhi
41.05
Year_To_Date Sales
186.09K
Maximum Profit Margin by Sub-Category
Furnishings
Kurti
Skirt
Trousers
Accessories
Stole
Saree
T-shirt
Bookcases
Shirt
Printers
Hankerchief
Electronic Games
Leggings
Phones
Chairs
4.88%
6.69%
5.21%
6.69%
5.47%
6.69%
6.14%
6.68%
6.17%
6.65%
6.29%
6.64%
6.33%
6.56%
6.42%
6.51%
Sales Target Achievement by Category
200K
150K
100K
50K
0K
135K
174K
135K
129K
135K
133K
Clothing
Electronics
Furniture
Actual Sales
Sales Target
Monthly Sales Trend
80K
60K
40K
20K
73K
78K
35K
97K
44K
24K
30K
53K
56K
82K
48K
31K
January
February
March
April
May
June
July
August
September
October
November
December
E-Commerce Sales Analysis
Comparison of Profit & Qty by Sub_Category
Quantity Sold
300
250
200
150
100
50
0
-4K
-2K
0K
2K
4K
Profit
Sales Target
436K
Revenue
641.04K